

DATA ANALYSIS

USING SQL & PYTHON

- *ANALYZE SALES PERFORMANCE*
- *IDENTIFY TOP-SELLING PRODUCTS*
- *UNDERSTAND CUSTOMER PURCHASING BEHAVIOR*
- *TRACK REVENUE TRENDS*
- *EVALUATE PAYMENT METHODS AND ORDER STATUSES*



Shop now

TABLE COLUMNS

Sheet1
CouponCode
CustomerID
Σ Date
Σ ItemsInCart
OrderID
OrderStatus
PaymentMethod
Product
Σ Quantity
ReferralSource
ShippingAddress
Σ TotalPrice
TrackingNumber
Σ UnitPrice

1.Display orders where TotalPrice is between 500 and 2000

```
1 -- Display orders where TotalPrice is between 500 and 2000.
2
3 • select * from e_commerce.ecom_order
4 where TotalPrice between 500 and 2000
```

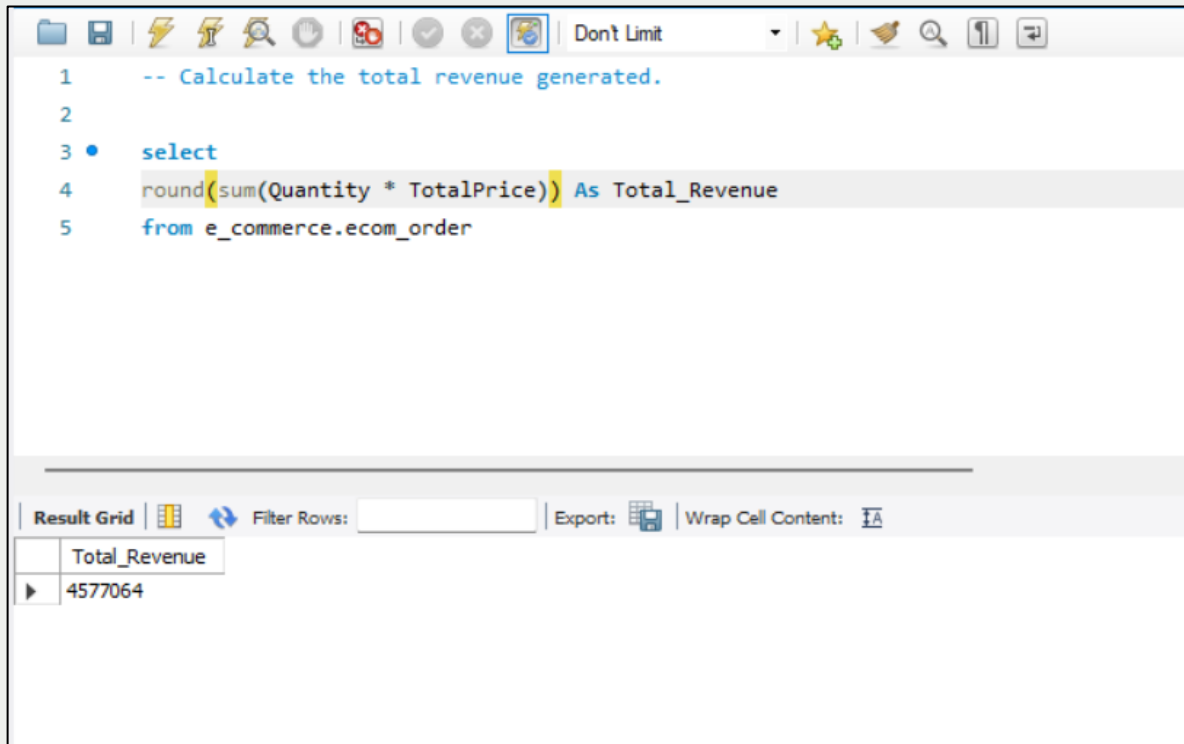
OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode
ORD200006	2025-06-17 00:00:00	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10
ORD200007	2023-05-12 00:00:00	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP
ORD200013	2023-08-30 00:00:00	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP
ORD200014	2023-03-27 00:00:00	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10
ORD200016	2023-01-24 00:00:00	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP
ORD200019	2023-01-23 00:00:00	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRKS59101365	9	SAVE10
ORD200021	2024-11-17 00:00:00	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRKS2646983	6	
ORD200023	2025-04-09 00:00:00	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP
ORD200028	2023-11-21 00:00:00	C59823	Phone	4	160.07	568 Main St	Online	Pending	TRK65895539	5	
ORD200029	2024-09-20 00:00:00	C33260	Desk	3	423.53	930 Main St	Credit Card	Delivered	TRKS9744090	6	SAVE10
ORD200030	2025-03-22 00:00:00	C44267	Monitor	2	426.7	895 Main St	Online	Pending	TRKS4751444	5	SAVE10
ORD200031	2023-06-30 00:00:00	C26445	Chair	3	500.08	996 Main St	Debit Card	Returned	TRK31229981	6	

2.Find orders where OrderStatus is either 'Delivered' or 'Shipped'.

```
1 -- Find orders where OrderStatus is either 'Delivered' or 'Shipped'
2
3 • select * from e_commerce.ecom_order
4 where OrderStatus = 'Delivered' or OrderStatus = 'Shipped'
```

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode
ORD200000	2023-01-04 00:00:00	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10
ORD200001	2024-08-23 00:00:00	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10
ORD200004	2025-05-08 00:00:00	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10
ORD200005	2023-10-23 00:00:00	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10
ORD200007	2023-05-12 00:00:00	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP
ORD200009	2023-11-21 00:00:00	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10
ORD200015	2023-07-17 00:00:00	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRKS4930938	3	
ORD200017	2024-03-02 00:00:00	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116	3	
ORD200021	2024-11-17 00:00:00	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRKS2646983	6	
ORD200024	2024-10-30 00:00:00	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15
ORD200025	2024-07-24 00:00:00	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15
ORD200026	2024-03-06 00:00:00	C21944	Laotoo	5	59.55	296 Main St	Online	Delivered	TRK80276812	6	

3. Calculate the total revenue generated.



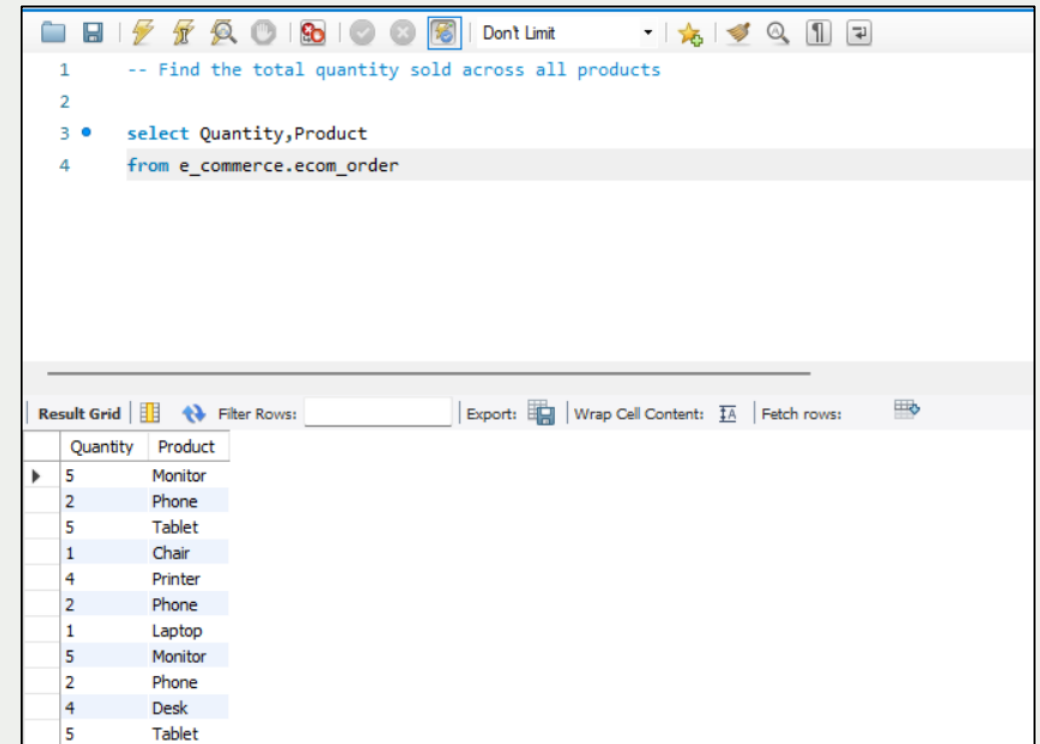
The screenshot shows a SQL IDE window with a toolbar at the top. The query editor contains the following SQL code:

```
1  -- Calculate the total revenue generated.
2
3  •  select
4     round(sum(Quantity * TotalPrice)) As Total_Revenue
5     from e_commerce.ecom_order
```

Below the query editor, the 'Result Grid' tab is active, displaying a single row of results:

Total_Revenue
4577064

4. Find the total quantity sold across all products..



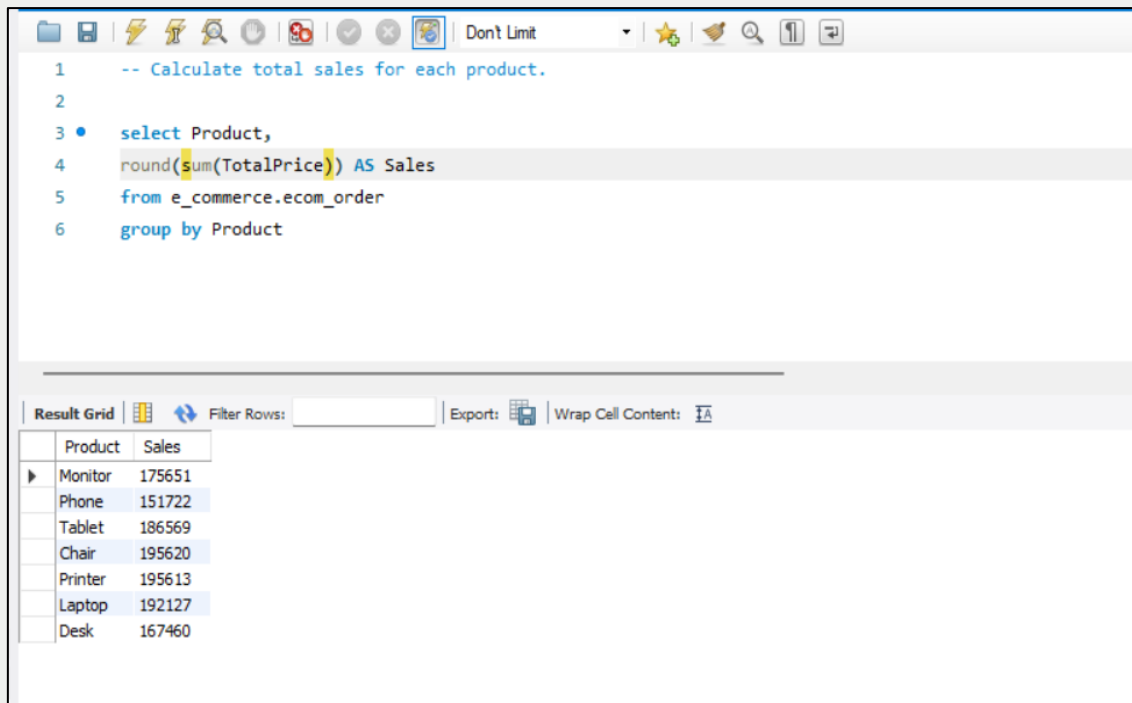
The screenshot shows a SQL IDE window with a toolbar at the top. The query editor contains the following SQL code:

```
1  -- Find the total quantity sold across all products
2
3  •  select Quantity, Product
4     from e_commerce.ecom_order
```

Below the query editor, the 'Result Grid' tab is active, displaying a list of products and their quantities:

Quantity	Product
5	Monitor
2	Phone
5	Tablet
1	Chair
4	Printer
2	Phone
1	Laptop
5	Monitor
2	Phone
4	Desk
5	Tablet

5. Calculate total sales for each product..



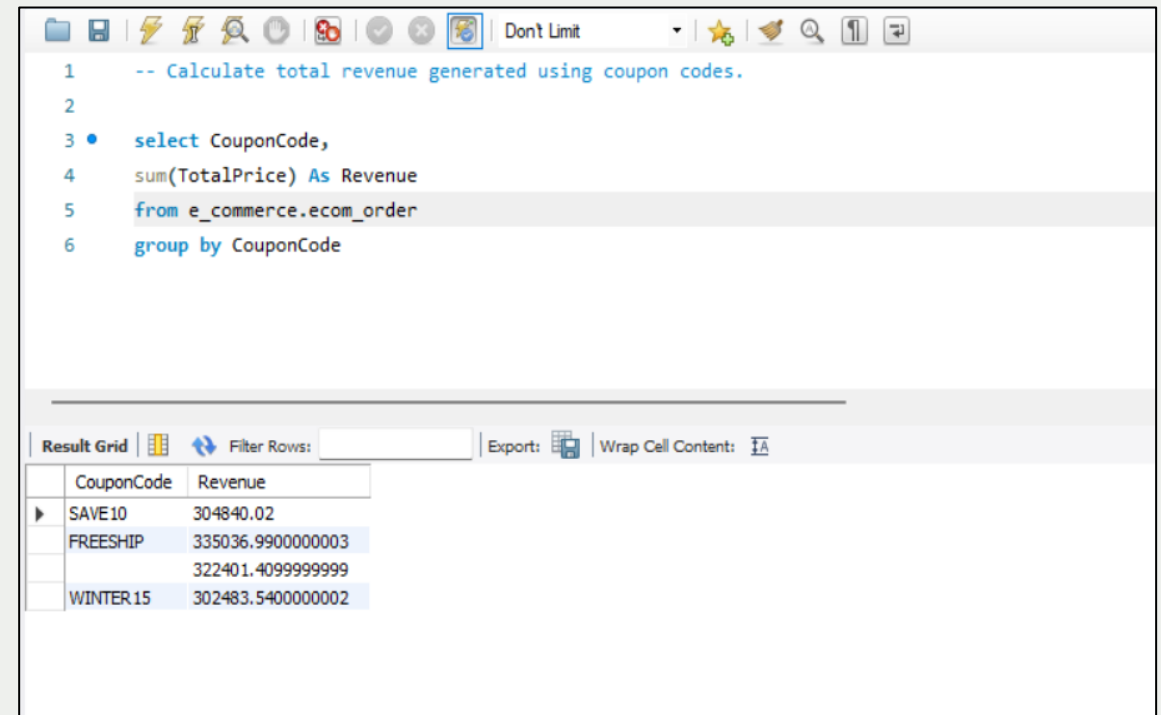
The screenshot shows a SQL IDE window with a toolbar at the top. The query editor contains the following SQL code:

```
1  -- Calculate total sales for each product.
2
3  •  select Product,
4     round(sum(TotalPrice)) AS Sales
5  from e_commerce.ecom_order
6  group by Product
```

Below the query editor is a 'Result Grid' section with a table showing the results of the query. The table has two columns: 'Product' and 'Sales'. The data is as follows:

Product	Sales
Monitor	175651
Phone	151722
Tablet	186569
Chair	195620
Printer	195613
Laptop	192127
Desk	167460

6. Calculate total revenue generated using coupon codes.



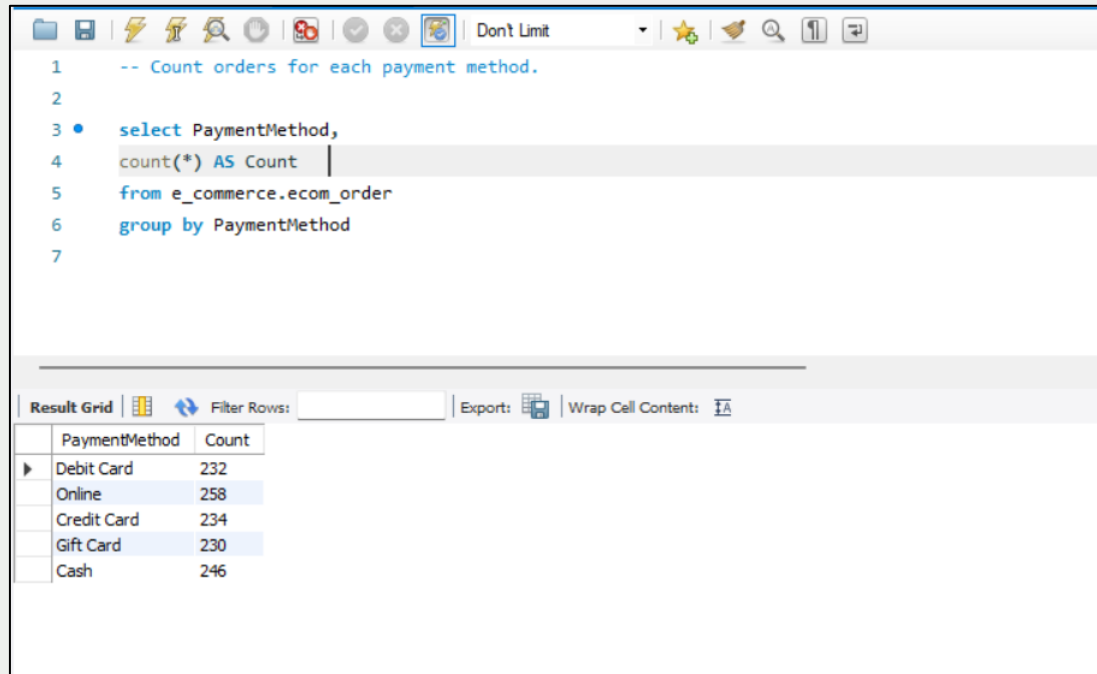
The screenshot shows a SQL IDE window with a toolbar at the top. The query editor contains the following SQL code:

```
1  -- Calculate total revenue generated using coupon codes.
2
3  •  select CouponCode,
4     sum(TotalPrice) As Revenue
5  from e_commerce.ecom_order
6  group by CouponCode
```

Below the query editor is a 'Result Grid' section with a table showing the results of the query. The table has two columns: 'CouponCode' and 'Revenue'. The data is as follows:

CouponCode	Revenue
SAVE10	304840.02
FREESHIP	335036.9900000003
	322401.40999999999
WINTER15	302483.5400000002

7.Count orders for each payment method

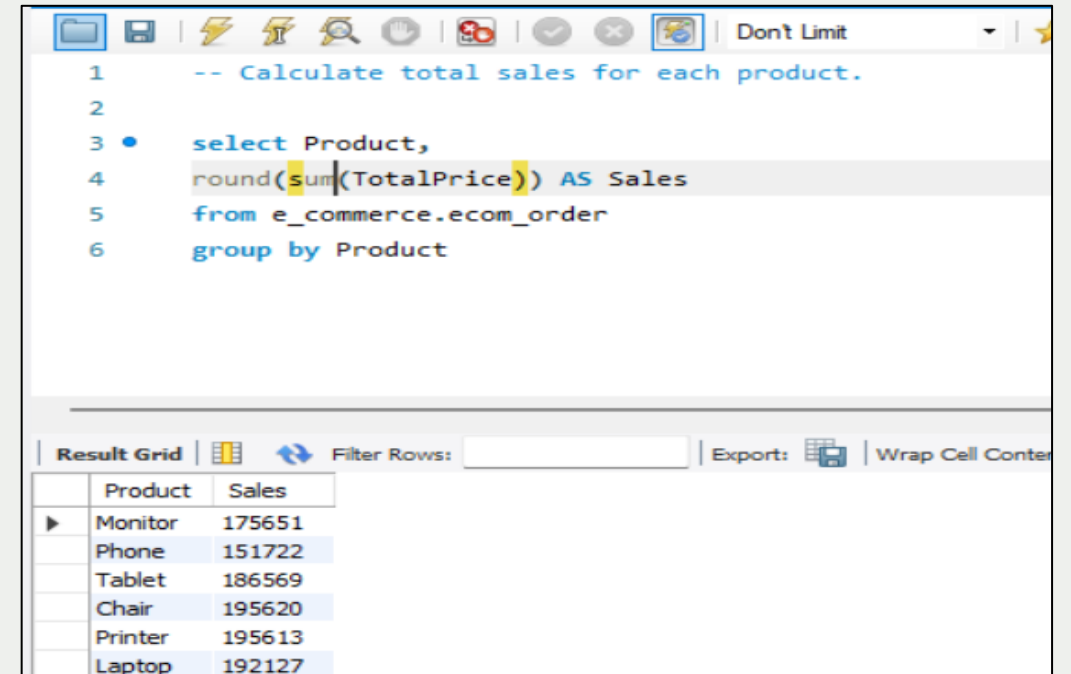


```
1  -- Count orders for each payment method.
2
3  • select PaymentMethod,
4     count(*) AS Count
5  from e_commerce.ecom_order
6  group by PaymentMethod
7
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	PaymentMethod	Count
▶	Debit Card	232
	Online	258
	Credit Card	234
	Gift Card	230
	Cash	246

8.Calculate total sales for each product.

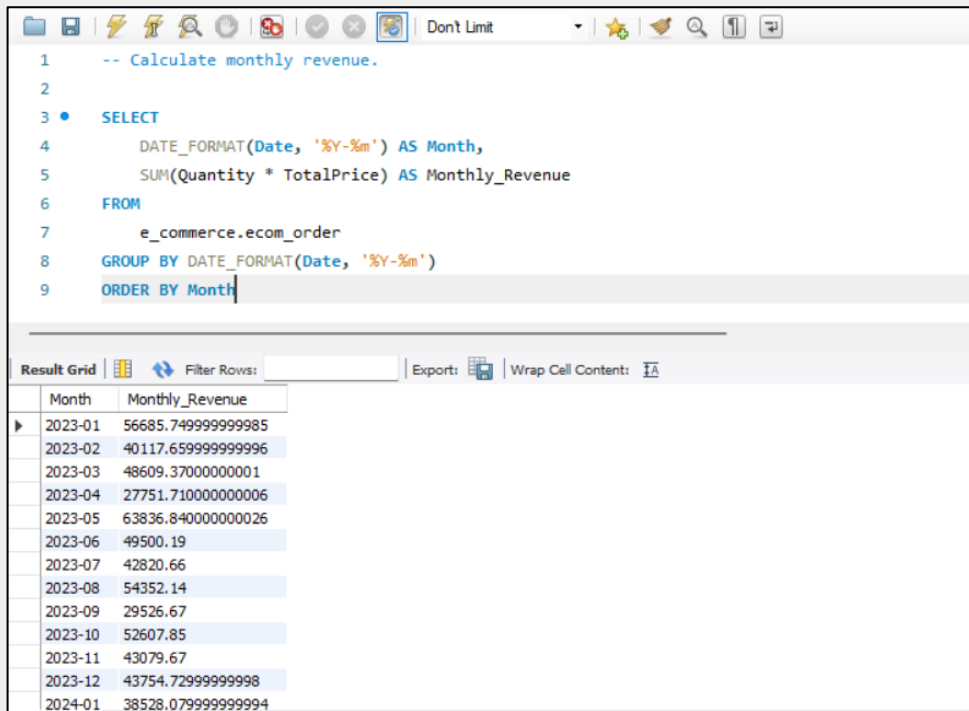


```
1  -- Calculate total sales for each product.
2
3  • select Product,
4     round(sum(TotalPrice)) AS Sales
5  from e_commerce.ecom_order
6  group by Product
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Product	Sales
▶	Monitor	175651
	Phone	151722
	Tablet	186569
	Chair	195620
	Printer	195613
	Laptop	192127

9. Calculate monthly revenue.

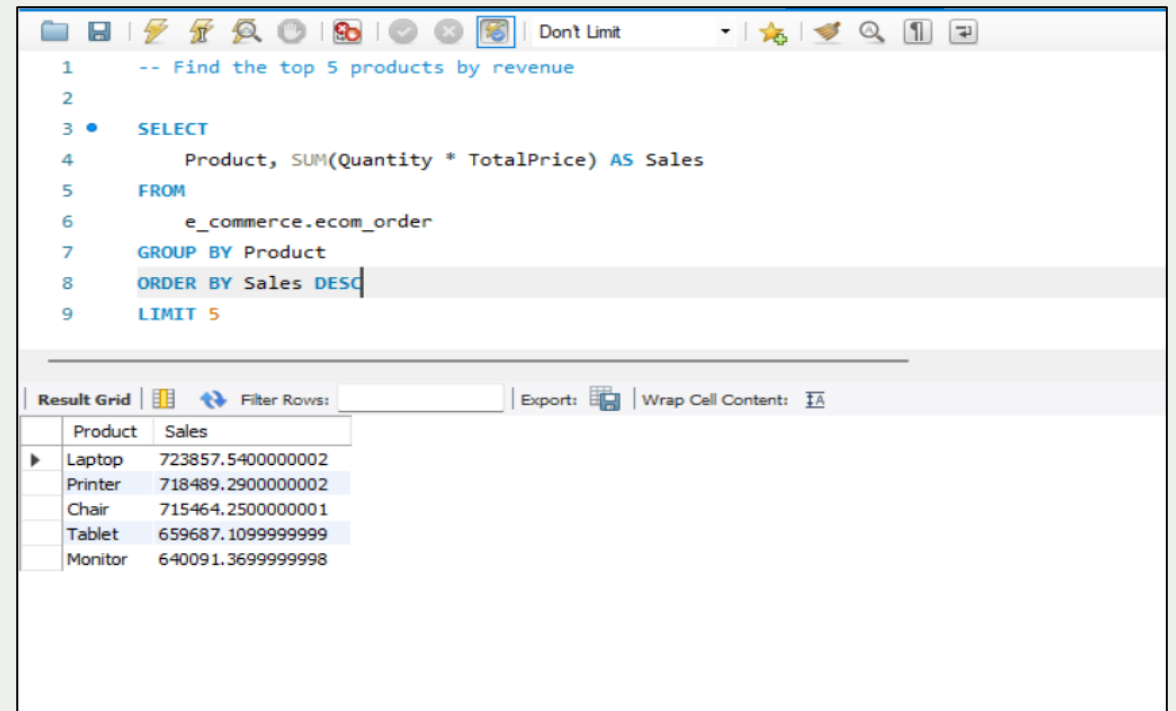


```
1  -- Calculate monthly revenue.
2
3  • SELECT
4      DATE_FORMAT(Date, '%Y-%m') AS Month,
5      SUM(Quantity * TotalPrice) AS Monthly_Revenue
6  FROM
7      e_commerce.ecom_order
8  GROUP BY DATE_FORMAT(Date, '%Y-%m')
9  ORDER BY Month
```

Result Grid

	Month	Monthly_Revenue
▶	2023-01	56685.749999999985
	2023-02	40117.659999999996
	2023-03	48609.370000000001
	2023-04	27751.710000000006
	2023-05	63836.8400000000026
	2023-06	49500.19
	2023-07	42820.66
	2023-08	54352.14
	2023-09	29526.67
	2023-10	52607.85
	2023-11	43079.67
	2023-12	43754.729999999998
	2024-01	38528.079999999994

10. Find the top 5 products by revenue.



```
1  -- Find the top 5 products by revenue
2
3  • SELECT
4      Product, SUM(Quantity * TotalPrice) AS Sales
5  FROM
6      e_commerce.ecom_order
7  GROUP BY Product
8  ORDER BY Sales DESC
9  LIMIT 5
```

Result Grid

	Product	Sales
▶	Laptop	723857.5400000002
	Printer	718489.2900000002
	Chair	715464.2500000001
	Tablet	659687.1099999999
	Monitor	640091.3699999998

11. Calculate cumulative revenue over time.

```
1  -- Calculate cumulative revenue over time.
2
3  • select Date,
4      sum(revenue) over(order by Date) as Cum_revenue
5  from
6  (select Date,
7      round(sum(Quantity*TotalPrice)) as revenue
8  from e_commerce.ecom_order
9  group by Date) as sales
```

Date	Cum_revenue
2023-01-01 00:00:00	3313
2023-01-02 00:00:00	6984
2023-01-03 00:00:00	8779
2023-01-04 00:00:00	24597
2023-01-05 00:00:00	42809
2023-01-06 00:00:00	47382
2023-01-07 00:00:00	56698
2023-01-08 00:00:00	69817
2023-01-10 00:00:00	73802
2023-01-11 00:00:00	82112
2023-01-12 00:00:00	98209

12. Rank products based on total quantity sold.

```
1  -- Rank products based on total quantity sold.
2
3  • select Product,
4      sum(Quantity) AS total_Sold,
5      rank () over(order by sum(Quantity) desc) AS product_rank
6  from e_commerce.ecom_order
7  group by Product
```

Product	total_Sold	product_rank
Chair	562	1
Printer	542	2
Laptop	535	3
Desk	508	4
Tablet	497	5
Monitor	480	6
Phone	411	7

13. Rank products within each order status by revenue.

```
1  -- Rank products within each order status by revenue.
2
3  • select OrderStatus,Product,
4    sum(TotalPrice) AS Revenue,
5    rank() over ( partition by OrderStatus order by sum(TotalPrice) desc) AS Product_Rank
6  from e_commerce.ecom_order
7  group by OrderStatus,Product
```

OrderStatus	Product	Revenue	Product_Rank
Cancelled	Chair	48660.979999999996	1
Cancelled	Laptop	43761.72	2
Cancelled	Tablet	40629.960000000001	3
Cancelled	Desk	39587.690000000001	4
Cancelled	Printer	38675.160000000001	5
Cancelled	Monitor	37057.44	6
Cancelled	Phone	28023.259999999995	7
Delivered	Laptop	40714.429999999998	1
Delivered	Phone	40345.41	2
Delivered	Printer	38054.73	3
Delivered	Monitor	35999.62	4
Delivered	Tablet	31794.519999999997	5
Delivered	Chair	31465.829999999998	6

14. Find the percentage contribution of each product to total revenue.

```
1  -- Find the percentage contribution of each product to total revenue.
2  • SELECT Product,
3    SUM(TotalPrice) AS ProductRevenue,
4    ROUND(
5      (SUM(TotalPrice) * 100.0) /
6      (SELECT SUM(TotalPrice) FROM e_commerce.ecom_order),
7      2
8    ) AS RevenuePercentage
9  FROM e_commerce.ecom_order
10 GROUP BY Product
11 ORDER BY RevenuePercentage DESC;
```

Product	ProductRevenue	RevenuePercentage
Chair	195620.110000000002	15.47
Printer	195612.609999999996	15.47
Laptop	192126.559999999994	15.19
Tablet	186568.950000000007	14.75
Monitor	175651.409999999997	13.89
Desk	167459.93	13.24
Phone	151722.39	12

15. Calculate each customer's percentage contribution to total revenue.

```
1  -- calculate each customer's percentage contribution to total revenue.
2
3  • SELECT
4      CustomerId,
5      SUM(TotalPrice) AS Revenue,
6      ROUND((SUM(TotalPrice) * 100.0) / (SELECT
7          SUM(TotalPrice)
8          FROM
9              e_commerce.ecom_order),
10         2) AS per_contr
11  FROM
12      e_commerce.ecom_order
13  GROUP BY CustomerId
14  ORDER BY Per_contr DESC
```

CustomerId	Revenue	per_contr
C38840	5723.23	0.45
C16775	3353.75	0.27
C65986	3352.4	0.27
C18404	3370.2	0.27
C13877	3384.9	0.27
C67260	3390.8	0.27
C57276	3456.4	0.27
C53464	3299.25	0.26
C13108	3293.85	0.26

16. Count orders for each payment method.

```
1  -- Count orders for each payment method.
2
3  • select PaymentMethod,
4      count(*) AS Count
5  from e_commerce.ecom_order
6  group by PaymentMethod
7
```

PaymentMethod	Count
Debit Card	232
Online	258
Credit Card	234
Gift Card	230
Cash	246

PANDAS

Fill Null values In 0

```
[19]: df=df.fillna(0)
```

```
[20]: df.isnull().sum()
```

```
[20]: OrderID      0
      Date        0
      CustomerID  0
      Product     0
      Quantity    0
      UnitPrice   0
      ShippingAddress 0
      PaymentMethod 0
      OrderStatus 0
      TrackingNumber 0
      ItemsInCart 0
      CouponCode  0
      ReferralSource 0
      TotalPrice  0
      dtype: int64
```

```
[21]: df.describe()
```

```
[21]:
```

	Quantity	UnitPrice	ItemsInCart	TotalPrice
count	1200.000000	1200.000000	1200.000000	1200.000000
mean	2.945833	356.412750	5.485000	1053.968300
std	1.407557	197.177146	2.281983	819.856558
min	1.000000	11.390000	1.000000	11.390000
25%	2.000000	186.062500	4.000000	410.520000
50%	3.000000	364.210000	5.000000	823.615000
75%	4.000000	521.570000	7.000000	1578.475000
max	5.000000	699.930000	10.000000	3456.400000

```
[8]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   OrderID                1200 non-null  object  
1   Date                   1200 non-null  object  
2   CustomerID             1200 non-null  object  
3   Product                1200 non-null  object  
4   Quantity               1200 non-null  int64   
5   UnitPrice              1200 non-null  float64  
6   ShippingAddress        1200 non-null  object  
7   PaymentMethod          1200 non-null  object  
8   OrderStatus            1200 non-null  object  
9   TrackingNumber         1200 non-null  object  
10  ItemsInCart            1200 non-null  int64   
11  CouponCode             891 non-null   object  
12  ReferralSource         1200 non-null  object  
13  TotalPrice             1200 non-null  float64  
dtypes: float64(2), int64(2), object(10)
memory usage: 131.4+ KB
```

```
[11]: df.shape
```

```
[11]: (1200, 14)
```

```
[18]: df.isnull().
```

```
[18]: OrderID      0
      Date        0
      CustomerID  0
      Product     0
      Quantity    0
      UnitPrice   0
      ShippingAddress 0
      PaymentMethod 0
      OrderStatus 0
      TrackingNumber 0
      ItemsInCart 0
      CouponCode 309
      ReferralSource 0
      TotalPrice  0
      dtvpe: int64
```

Total Revenue

```
[23]: df["TotalPrice"].sum()
```

```
[23]: 1264761.96
```

```
[24]: df["TotalPrice"].mean()
```

```
[24]: 1053.9683
```

```
[27]: df.groupby("Product")["Quantity"].sum().sort_values(ascending=False).head(5)
```

```
[27]: Product
Chair    562
Printer  542
Laptop   535
Desk     508
Tablet   497
Name: Quantity, dtype: int64
```

```
[35]: df[df['OrderStatus'] == 'Delivered']
```

```
[35]:
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Coupon	
	4	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	S
	15	ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3	
	21	ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6	
	24	ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WIN
	25	ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WIN
	...	...	...	...	...	...	...	...	...	...	...	...	...

```
[37]: df.loc[df['TotalPrice'].idxmax()]
```

```
[37]: OrderID      ORD200789
Date          2023-08-17
CustomerID    C57276
Product       Tablet
Quantity      5
UnitPrice     691.28
ShippingAddress 183 Main St
PaymentMethod  Online
OrderStatus    Delivered
TrackingNumber TRK75899752
ItemsInCart    10
CouponCode     SAVE10
ReferralSource Email
TotalPrice     3456.4
Name: 789, dtype: object
```

```
[38]: df.groupby("CustomerID")["TotalPrice"].sum().sort_values(ascending=False).head(10)
```

```
[38]: CustomerID
C38840    5723.23
C57276    3456.40
C67260    3390.80
C13877    3384.90
C18404    3370.20
C16775    3353.75
C65986    3352.40
C47778    3334.00
C59183    3322.55
C25276    3313.90
Name: TotalPrice, dtype: float64
```

```
[42]: df[df['PaymentMethod']=='Credit Card'].value_counts()
```

```
[42]:
```

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Coupon
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FR
EESHIP	Email	2753.40	1								
ORD200859	2024-08-28	C11127	Chair	5	261.33	154 Main St	Credit Card	Returned	TRK73285507	5	SA
VE10	Referral	1306.65	1								
ORD200816	2024-02-21	C55286	Tablet	5	574.79	497 Main St	Credit Card	Pending	TRK39058734	8	WI

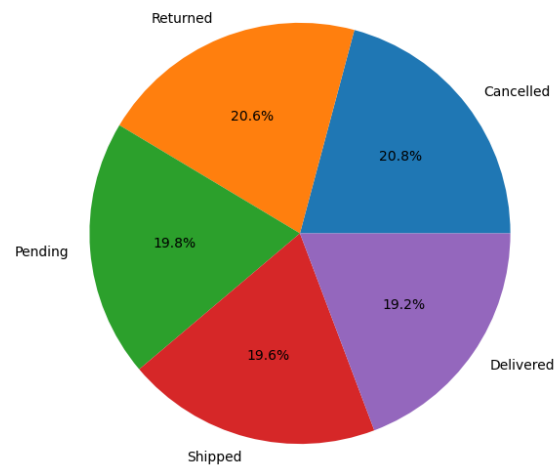
VISUALIZATUIN USING MATPLOTLIB

```
[62]: import matplotlib.pyplot as plt

status_counts = df["OrderStatus"].value_counts()

plt.figure(figsize=(7, 7))
plt.pie(
    status_counts,
    labels=status_counts.index,
    autopct="%1.1f%%"
)
plt.title("Order Status Distribution")
plt.show()
```

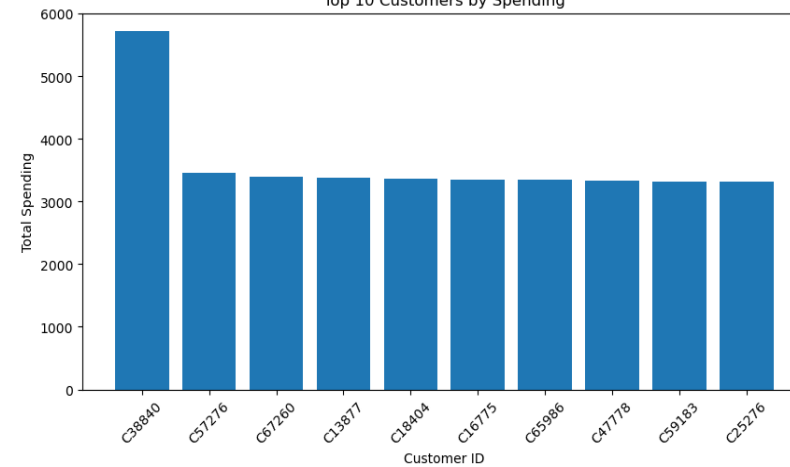
Order Status Distribution



```
[64]: customer_sales = (
    df.groupby("CustomerID")["TotalPrice"]
      .sum()
      .sort_values(ascending=False)
      .head(10)
)

plt.figure(figsize=(10, 5))
plt.bar(customer_sales.index.astype(str), customer_sales.values)
plt.title("Top 10 Customers by Spending")
plt.xlabel("Customer ID")
plt.ylabel("Total Spending")
plt.xticks(rotation=45)
plt.show()
```

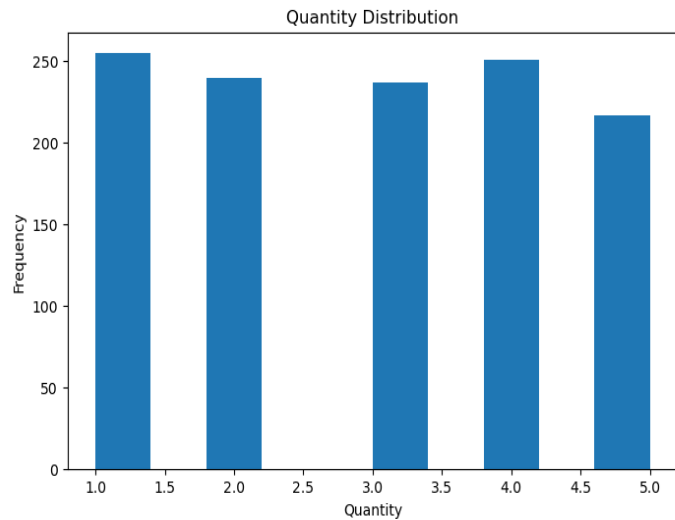
Top 10 Customers by Spending



VISUALIZATUIN USING MATPLOTLIB

Customer ID

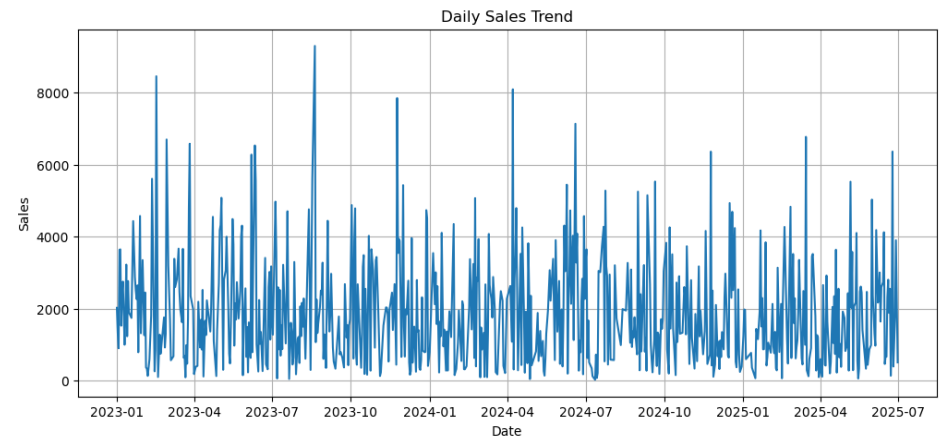
```
[65]: plt.figure(figsize=(8, 5))
plt.hist(df["Quantity"], bins=10)
plt.title("Quantity Distribution")
plt.xlabel("Quantity")
plt.ylabel("Frequency")
plt.show()
```



```
[63]: df["Date"] = pd.to_datetime(df["Date"])

daily_sales = df.groupby("Date")["TotalPrice"].sum()

plt.figure(figsize=(12, 5))
plt.plot(daily_sales.index, daily_sales.values)
plt.title("Daily Sales Trend")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.grid(True)
plt.show()
```



Thank you

SACHIN CHAVAN

9623074077

sachuchavan81@gmail.com